

# Hossam Yasser Ibrahim Esmail

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## SUMMARY

Infrastructure Design & BIM Engineer with experience in infrastructure engineering, BIM implementation, and digital delivery of utility networks and road infrastructure projects. Proficient in Autodesk Civil 3D, AutoCAD, Navisworks, Autodesk Inventor, and Dynamo for BIM modeling, design automation, parametric modeling, LOD 400/450 development, clash detection, and multidisciplinary BIM coordination. Experienced in delivering infrastructure design packages, including BODR, Schematic Design (SD), Tender Design (TD), Detailed Design (DD), shop drawings, construction documentation, and BIM deliverables. Skilled in the design of water supply, sewerage, stormwater drainage, firefighting, and irrigation networks using WaterCAD, WaterGEMS, SewerCAD, and SewerGEMS. Strong expertise in Civil 3D customization, Infrastructure Parts Editor (IPE), Catalog Part Content, Subassembly Composer, and development of customized parametric infrastructure components using Autodesk Inventor. Experienced in BIM coordination, utility modeling, design automation, hydrology analysis, flood risk assessment, and delivering construction-ready BIM models.

## WORK EXPERIENCE

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### **DMA for Design and Engineering Consultancy – Sheraton, Egypt**

#### ***Infrastructure Design & BIM Engineer***

[01/2026-Present]

- Develop infrastructure network designs, including water supply, sewerage, stormwater, and utility systems.
- Prepare Basis of Design Reports (BODR) and deliver Schematic Design (SD) and Tender Design (TD) packages.
- Coordinate multidisciplinary designs with Plumbing, Fire Protection, and Electrical teams to ensure fully coordinated project deliverables.
- Develop BIM models for infrastructure networks using Autodesk Civil 3D.
- Automate engineering workflows using Dynamo to improve productivity and reduce repetitive tasks.
- Create and customize Civil 3D components, including Manholes, Valves, and other utility objects, using Autodesk Inventor, Infrastructure Parts Editor (IPE), Catalog Part Content, and Subassembly Composer.
- Contribute to the development of BIM standards and optimized workflows for infrastructure projects.

### **IMAGINE CONSULTANT ENGINEERING – New Cairo, Egypt**

#### ***Infrastructure Design & BIM Engineer***

[10/2024-01/2026]

- Modeled complete infrastructure utility networks (sewage, water supply, stormwater, firefighting, grey water, medium voltage, low voltage, and light current) using Autodesk Civil 3D.
- Performed clash detection and coordination across multiple networks with Navisworks, delivering conflict-free and constructible designs.
- Prepared high-quality, fully coordinated shop drawings for utility networks and road works, ensuring efficient and accurate site execution.
- Contributed to road shop drawings for the Jazan City for Primary and Downstream Industries (JCPDI), Saudi Arabia, gaining advanced experience in road detailing and coordination.
- Collaborated with design and site teams to resolve technical issues and improve constructability.
- Assisted in the design of infrastructure networks (sewage, water supply, stormwater, irrigation, and firefighting), strengthening both technical accuracy and practical expertise.

### **United Engineers Contracting Company – Ninth District, October City**

#### ***Site ENGINEER TRAINEE***

[07/2022-08/2022]

- Gained hands-on site experience in carpentry, reinforcement works, formwork installation, and concrete casting for structural elements, developing a solid understanding of construction processes.

### **Trainee for CMB Company– Giza, Egypt.**

#### ***Civil ENGINEER TRAINEE***

[02/2022-02/2022]

- Acquired hands-on experience with modern construction chemicals, supporting quality control and learning about advanced materials and their applications in construction projects.

- Performed construction site operations and management, assisted in structural design implementation, ensured quality control, and collaborated with engineering teams to achieve project milestones

## Key PROJECTS

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**Mountain View – Creek View Compound – New Cairo, Egypt** [07/2026–Present]

Designed and coordinated water supply, firefighting, irrigation, and sewerage networks, preparing BODR, SD, and TD packages while ensuring multidisciplinary coordination and compliance with project standards.

**Urove Compound– New Zayed, Egypt** [06/2026–07/2026]

Designed water, sewer, stormwater, and irrigation networks, prepared BODR, SD, TD, BOQs, and design documentation, and coordinated multidisciplinary teams to deliver integrated infrastructure designs.

**Mountain View – Kingsway– 6<sup>th</sup> October, Egypt** [05/2026-7/2026]

Developed Civil 3D models for wet and dry utility networks, coordinated multidisciplinary BIM models, performed clash detection, and delivered coordinated 3D models and BIM deliverables.

**Mountain View – ALIVA – Phase 02– El Mostakbal City, Egypt** [02/2026-04/2026]

Designed potable water, sewer, stormwater, and irrigation networks, performed hydraulic calculations and network sizing, and coordinated multidisciplinary utility designs while preparing infrastructure design deliverables.

**Mountain View – Zone C (Evia), Ras El Hekma, Egypt** [01/2026–03/2026]

Designed potable water, sewer, stormwater, and irrigation networks, performed hydraulic calculations, pipe sizing, and design checks, and coordinated multidisciplinary teams to deliver integrated infrastructure designs.

**HRS High-Speed Rail Project – Helwan, Egypt** [12/2025-01/2026]

Performed BIM modeling and coordination of dry and wet utility networks using Civil 3D and Navisworks, and developed LOD 400 utility parts using Autodesk Inventor.

**AL NAEEM Residential Compound – Saudi Arabia** [11/2025-12/2025]

Designed complete water supply and sanitary sewer networks, including network layout planning, pipe sizing, and hydraulic analysis using WaterCAD, SewerCAD, and InfraWizard, ensuring compliance with project requirements and Saudi standards.

**Rose Ville Project– Cairo, Egypt** [08/2025-11/2025]

Designed combined sewage and stormwater networks, water supply and firefighting networks, and irrigation systems using SewerCAD and WaterCAD, enhancing infrastructure planning and design expertise.

**Belle Vie Sports Club – New Zayed, Egypt** [05/2025-11/2025]

Performed BIM modeling and coordination of infrastructure and utility networks (water supply, sewage, firefighting, medium voltage, low voltage, and telecom systems) using Civil 3D and Navisworks, delivering LOD 400 shop drawings and creating composite coordination panels to ensure conflict-free and efficient construction execution.

**Sewage Network Design for Katamya, Egypt** [03/2025-05/2025]

Designed complete sanitary sewer systems, including layout planning, hydraulic calculations, and pipe sizing, and prepared detailed drawings using InfraWizard and SewerGEMS.

**Sewage Network Design for Attika – New Capital, Egypt** [04/2025-05/2025]

Performed comprehensive hydraulic analysis of sanitary sewer networks, assessing system performance, identifying bottlenecks, and proposing design modifications to improve efficiency and reliability.

**Jazan City for Primary and Downstream Industries (JCPDI) – Saudi Arabia** [12/2024-06/2025]

Involved in BIM modeling and coordination of utility and road networks, preparing clash-free models and shop drawings, including roads shop drawings, in addition to creating composite coordination panels for seamless integration.

**Al Kharj Entertainment Complex – Saudi Arabia** [10/2024-08/2025]

Performed BIM modeling and coordination of infrastructure and utility networks (sewage, water supply, stormwater, firefighting, grey water, medium voltage, low voltage, and light current systems) using Civil 3D, Navisworks, and Inventor, delivering LOD 400 fully coordinated shop drawings for construction.

# TECHNICAL SKILLS

## Proficient In:

- Autodesk Civil 3D
- Autodesk Navisworks
- Autodesk Inventor
- Autodesk AutoCAD
- Infrastructure Parts Editor (IPE)
- SewerCAD
- SewerGEMS
- WaterCAD
- WaterGEMS
- Catalog Part Content
- InfraWizard
- Microsoft Office (Word, Excel, PowerPoint)
- Dynamo
- Subassembly Composer

## Familiar With:

- Autodesk Revit
- WMS (Watershed Modeling System)
- HEC-HMS
- HEC-RAS
- ArcGIS
- Global Mapper
- Culvert Master
- Flow Master
- CSI SAP2000
- SAFE
- ETABS
- CSI Column

# EDUCATION

## Bachelor’s Degree in Civil Engineering, Cairo University – Faculty of Engineering

- Grade: Very Good
- Graduation Project: Excellent

## Graduation Project: Adaptive Infrastructure Design: A Holistic Approach to Marsa Alam’s Flood

Resilience [02/2024 – 07/2024]

- Engineered a comprehensive flood protection strategy, integrating external measures (dams, canals, ponds) with internal stormwater networks.
- Applied ArcGIS, WMS, HEC-HMS, HEC-RAS, Civil 3D, Culvert Master, Flow Master, and SewerGEMS for hydrologic/hydraulic analysis, protection structure design, and water flow modeling.
- Demonstrated the ability to integrate diverse engineering solutions into an effective flood management system, showcasing advanced skills in flood risk mitigation and civil design.

# Courses & Certifications

- Comprehensive Utility Networks Design Course: Covered design of all infrastructure networks (sewage, water supply, stormwater, firefighting, irrigation) with understanding of network details and specifications, using SewerCAD, SewerGEMS, WaterCAD, WaterGEMS, InfraWizard, and Civil 3D.
- Road & Utility Coordination Course: Focused on preparing road shop drawings integrated with utility networks, ensuring full coordination and clash-free design.
- Structural Design Coursework: Skilled in AutoCAD 2D, CSI SAP2000, SAFE, ETABS, and Column, with foundational Autodesk Revit skills for BIM and building design.
- Advanced SAP Structural Analysis, Civil Academy: Covered load analysis, structural behavior, and advanced design techniques to enhance structural engineering proficiency.
- AutoCAD 2D Certification: Completed full-scale civil and architectural projects, gaining practical 2D drafting experience.
- Autodesk Revit Basics: Developed skills in building modeling, family creation, and project management for architectural and structural designs.

# LANGUAGE SKILLS

Arabic: Native language English: Proficient (written & spoken)